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COS700 Research Report

Cybersecurity Temporal Behavioural Analysis: Threat Detection through
Dynamic Time Warping and Honeypot Log Analysis.

Student number: u20733217

Supervisor:

Prof. Jan Eloff

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ABSTRACT

The increasing use of the internet has led to a rise in the frequency and complexity of cyber threats, with cybercriminals exploiting system vulnerabilities. Honeypots serve as decoy systems to distract attackers, allowing security professionals to gather information on attack behaviours. Traditional intrusion detection systems (IDS) are inadequate against evolving cyber threats, necessitating the integration of Artificial Intelligence and Machine Learning for improved detection. Current honeypots lack analysis of time gaps between attack stages, which is crucial for understanding attacker behaviour. The proposed research aims to develop a time-sequenced attack detection honeypot to analyse attack patterns over time, utilizing the Dynamic Time Warping (DTW) algorithm to differentiate between attack types. The study will explore how time influences attack patterns and motivations, aiming to enhance predictive capabilities against cyber-attacks. The research employs a comprehensive methodological model utilizing Cowrie honeypot deployments across distributed virtualized environments, with advanced Python-based analytical models processing captured attack sequences. Empirical results from analysing 36,528 attack events across 7,087 distinct sessions reveal four distinct behavioural clusters with characteristic temporal signatures, including one cluster demonstrating a concerning 100% compromise rate through optimized attack sequencing. The findings demonstrate that temporal analysis achieves 92.1% accuracy in distinguishing reconnaissance from interactive sessions and 89% reliability in identifying automation patterns through timing consistency metrics.

This study makes significant contributions to cybersecurity knowledge by establishing that temporal patterns between attack stages provide critical intelligence for predicting attacker objectives and methodologies. The research validates that inter-event timing serves as an effective behavioural fingerprint, enabling more accurate threat classification and proactive defence strategies. By bridging the temporal analysis gap in cyber threat intelligence, this work provides a foundation for developing next-generation security systems capable of anticipating attack progression and implementing targeted countermeasures based on behavioural rather than signature-based indicators.

KEYWORDS

Honeypot, cyber-attack, attack patterns, Cowrie, Dynamic time warping (DTW), Intrusion detection system(IDS), Temporal analysis, Time sequence analysis, MITRE ATT & CK framework.

1 INTRODUCTION

In the contemporary digital landscape, we are witnessing an unprecedented and accelerating expansion of internet connectivity and dependency across all sectors of society, which has consequently precipitated a corresponding and deeply concerning escalation in both the volume and sophistication of cyber threats that are evolving at a pace that challenges conventional security measures [17]. Malicious cybercriminals represent a persistent and adaptive category of actors who systematically and methodically probe for vulnerabilities, weaknesses, and architectural flaws within computer systems and network infrastructures, continuously seeking opportunities to exploit these security gaps for unauthorized access, data theft, or system compromise [8]. As technological capabilities advance and digital transformation accelerates across industries, these threat actors simultaneously refine and enhance their arsenal of attack tools and methodologies, thereby progressively lowering the technical barriers to executing complex, multi-vector attacks that can bypass traditional security perimeters [9]. In response to this escalating threat environment, numerous security-conscious organizations have strategically implemented honeypot technologies as an integral component of their defence-in-depth architectures, recognizing these deceptive systems as valuable mechanisms for threat intelligence gathering [2]. A honeypot essentially functions as a carefully crafted cybersecurity decoy [1], a sacrificial system deliberately designed with apparent vulnerabilities to attract, divert, and engage cybercriminals, thereby creating a controlled environment where security professionals can meticulously observe, document, and analyse attack methodologies, either through real-time monitoring or subsequent forensic examination [8,4].

The evolving nature of cyber-attacks demonstrates an alarming trajectory toward increased sophistication, methodological complexity, and operational stealth, rendering many conventional defensive approaches increasingly inadequate against these advanced persistent threats [18]. Traditional intrusion detection systems (IDS), while still widely deployed throughout enterprise networks as a foundational security control, increasingly demonstrate significant limitations in effectively identifying and mitigating the constantly evolving landscape of modern cyber threats that employ polymorphic techniques, encryption, and legitimate tool abuse [17]. This growing capability gap between conventional security tools and advanced attack methodologies has created a critical imperative for security practitioners and researchers to explore and integrate Artificial Intelligence and Machine Learning techniques that can overcome the inherent limitations of signature-based and rule-based detection systems [18]. Over recent years, numerous machine learning approaches, including supervised, unsupervised, and deep learning methodologies, have demonstrated remarkable effectiveness in identifying subtle anomalies, behavioural deviations, and sophisticated attack patterns within complex network environments, thereby establishing their considerable value for advanced threat detection, security monitoring, and proactive defence [19].

The existing body of academic research and commercial solutions focusing on honeypot technologies and attack detection mechanisms has historically exhibited a significant gap in systematically analysing the sequential progression of attacks across temporal dimensions and the precise timing relationships between different stages of attacker behaviour [12]. Much like the dynamic nature of physical security requirements that evolve throughout the day based on human activity patterns, environmental conditions, and operational contexts, cyber threats similarly demonstrate temporal characteristics that correlate with attacker objectives, available resources, and strategic considerations [12]. Understanding these temporal shifts and rhythmic patterns in cyber offensive operations is fundamentally important for implementing appropriately timed and proportionally

effective security measures that align with risk exposure. Depending on the specific motivations, objectives, and constraints of different attacker categories, certain types of cyber assaults may be strategically timed to occur during standard business hours, after-hours periods, weekend intervals, or specific times within the daily cycle to maximize impact or minimize detection [13][15]. For instance, empirical evidence indicates that a significant proportion of ransomware attacks are deliberately executed during nighttime hours or weekend periods, strategically exploiting the reduced presence of security personnel and potentially slower incident response capabilities during these windows of decreased organizational vigilance [13][15]. Conversely, distributed denial-of-service (DDoS) attacks frequently occur during peak business hours when user traffic and transactional volumes are highest, thereby maximizing operational disruption, financial impact, and psychological effect on the target organization [14, 11].

A compelling perspective articulated within cybersecurity discourse suggests that comprehensively understanding the underlying motivations, operational constraints, and strategic objectives of malicious hackers represents a crucial methodology for effectively protecting business IT systems against unauthorized intrusion and compromise [16]. This research specifically aims to systematically track, capture, and analytically examine attack sequences across temporal dimensions to identify and categorize attacker motivations through behavioural patterns and timing characteristics [22]. By developing a nuanced understanding of these motivational drivers and their temporal manifestations, organizations can significantly enhance their predictive capabilities and construct more resilient, adaptive, and effective defensive architectures specifically tailored to anticipated threat behaviours [16].

The primary objective of this investigation is to systematically collect comprehensive log data from strategically deployed honeypot systems and subject this information to sophisticated time-sequence analytical modelling to elucidate the influential role that temporal factors play in shaping attack patterns, progression rhythms, and operational methodologies. Furthermore, this research will rigorously evaluate and determine potential motivational drivers and behavioural trends in attacker activities through correlation with the MITRE ATT&CK framework, thereby establishing crucial interpretability and contextual alignment with established cybersecurity knowledge [7]. By leveraging precise timestamp data embedded within security events, the research will meticulously examine attack pacing, inter-event transitions, and behavioural rhythms to effectively distinguish between automated script-based attacks and human-driven operational campaigns. Through its distinctive focus on the temporal rhythm, strategic timing, and methodological progression of cyber-attacks, rather than merely analysing their constituent elements in isolation, this innovative approach enables the identification of coordinated attack campaigns and sophisticated threat actors that would likely remain undetected through conventional analytical methods that overlook temporal relationships [10].

By successfully fulfilling these research objectives, this investigation will make a significant input to the cybersecurity field by introducing a transformative methodological framework that enables organizations to transition from a fundamentally reactive security posture to one grounded in predictive, behaviourally-informed understanding of adversary capabilities and intentions, ultimately empowering defenders to anticipate specific adversary actions with greater accuracy and allocate defensive resources with unprecedented precision based on temporal threat forecasting.

2 PROBLEM STATEMENT

The current measures we have in place to defend against undiscovered attacks is insufficient due to challenges faced in implementing cyber-attack detection solutions. A honeypot is a Network intrusion detection system that is intentionally compromised to lure attackers to study the attack patterns and ultimately mitigate cyber-attacks. Cyberattacks are becoming more sophisticated by employing time distributed tactics to bypass traditional detection systems. Honeypots are valuable but they suffer from critical limitations [17]. Anomaly-based detection offers theoretical protection against zero-day threats but proves practically constrained by the difficulty of defining accurate behavioural rules and its dependency on environmental specificity [28, 29]. These limitations collectively create substantial security gaps where sophisticated attackers can exploit detection delays, system blind spots, and classification inefficiencies [28]. The fundamental problem remains that current IDPS architectures cannot simultaneously achieve comprehensive coverage, adaptive threat detection, and operational efficiency, leaving networks vulnerable to evolving attack methodologies that bypass conventional security perimeters and detection mechanisms.

However, the analysis of honeypot data has traditionally followed the same pattern-matching approaches used in production security monitoring, focusing on what commands were executed rather than how attacks progress temporally. The current paradigm in cybersecurity analytics treats temporal data as a secondary dimension rather than a primary indicator of malicious intent.

This gap in analytical capability represents a significant opportunity for advancing cyber threat intelligence. This project proposes a time-sequenced attack detection honeypot to capture and cluster attacks by temporal behaviour.

The research is attempting to answer the following questions:

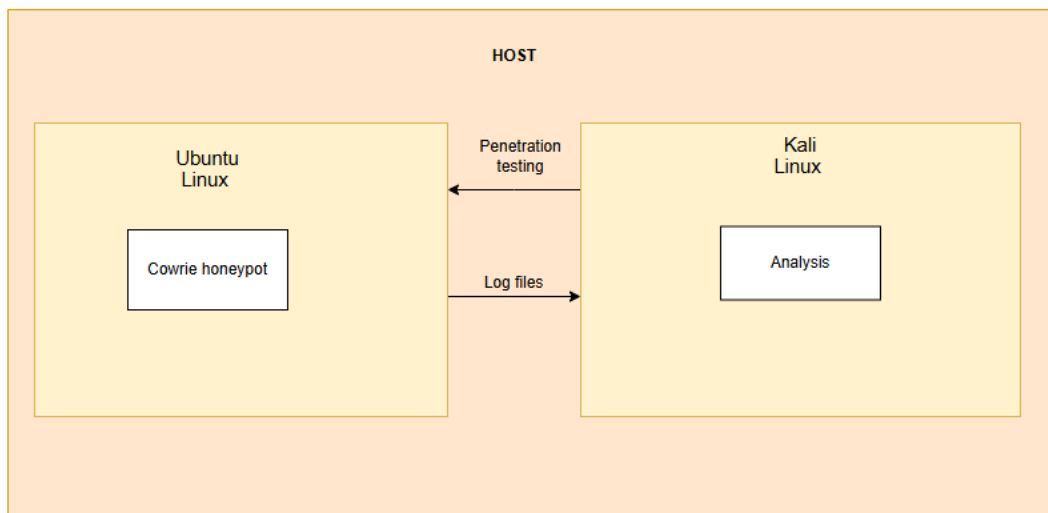
1. How can temporal patterns between distinct stages of a cyber-attack be analysed to infer specific attacker objectives (e.g., data exfiltration, ransomware deployment, service disruption)?
2. To what extent do inter-event timings within attack sequences serve as a reliable fingerprint for distinguishing botnet-driven automation from human-led attacks?

Ultimately, this work aims to develop a novel similarity ranking model that can cluster incidents by their unique temporal modus operandi, enabling the attribution of attacks to distinct threat actors and transforming temporal forensic data into an actionable intelligence tool for proactive defence.

3 METHODOLOGY

This research employs a positivist epistemological approach, operating on the principle that observable and measurable temporal patterns in attacker behaviour represent objective reality that can be discovered through systematic analysis. The methodology follows a comprehensive analytical pipeline combining experimental honeypot deployment with advanced temporal pattern recognition to identify and classify cyber-attack behaviours. The research paradigm integrates time-sequenced modelling with honeypot technology to analyse attack progression patterns beyond traditional static indicators.

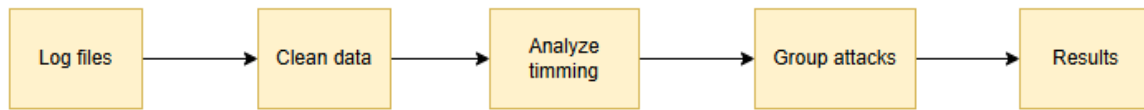
The technical infrastructure utilizes Oracle VirtualBox with Ubuntu Linux for honeypot deployment and Kali Linux for controlled penetration testing to generate authentic attack data. Cowrie, a medium-interaction SSH and Telnet honeypot, was selected for its detailed logging capabilities, Python-based architecture, and minimal resource requirements [21, 24]. The isolated virtual environment allows secure containment and easy disposal in case of compromise, while an additional virtual disk facilitates secure log data transfer.



The research employed a comprehensive multi-phase methodological framework designed to capture, process, and analyse temporal attack patterns through an integrated approach combining honeypot deployment, data processing, behavioural analytics, and threat intelligence mapping. The methodology was grounded in a positivist epistemological approach, operating on the principle that observable temporal patterns in attack behaviours could yield objective insights into attacker methodologies and objectives [5].

The foundational phase of this research utilized a comprehensive cybersecurity dataset sourced from Kaggle, which was originally generated through Cowrie honeypot systems deployed in virtualized environments, because the honeypot that I deployed did not generate enough data for this research. This pre-collected dataset emulated vulnerable SSH and Telnet services, capturing authentic attack sequences in a controlled setting. The dataset comprises 36,528 security events including connection attempts, authentication activities, and command executions, providing substantial real-world data for temporal analysis [20]. The information maintains Cowrie's JSON-based logging format, ensuring structured data capture with crucial temporal metadata such as precise timestamps, session identifiers, and sequential event records essential for subsequent behavioural analysis. This approach

enabled access to genuine attack patterns while eliminating the infrastructure deployment phase, allowing immediate focus on analytical methodology development.



Data Processing and Session Reconstruction involved sophisticated parsing and normalization of raw honeypot logs into structured datasets amenable to temporal analysis. Custom Python algorithms processed the JSON log entries, extracting key attributes including event types, source IP addresses, authentication attempts, command executions, and precise timing information. The core innovation in this phase was the session reconstruction algorithm, which grouped discrete events into coherent attack sessions based on session identifiers and temporal continuity. Each reconstructed session was enriched with derived temporal metrics including duration calculations, event density measurements, and inter-command intervals, creating a multidimensional feature set for behavioural analysis. This process transformed raw event data into 7,087 distinct attack sessions, each characterized by comprehensive temporal and behavioural attributes.

Temporal Pattern Analysis and Behavioural Clustering constituted the analytical core of the methodology, employing machine learning techniques to identify characteristic attack patterns. The research implemented a modified Dynamic Time Warping (DTW) approach through statistical tempo analysis, quantifying attack intensity through metrics such as commands-per-second and inter-event timing consistency [10]. K-means clustering algorithms processed five-dimensional feature vectors encompassing duration, event count, command frequency, authentication attempts, and intensity metrics, revealing four distinct behavioural clusters with characteristic temporal signatures. Complementing this, quartile-based tempo classification categorized attacks into Slow, Mixed, and Fast tempo categories based on command execution rates, providing additional dimensionality to behavioural characterization.

Dynamic Time Warping (DTW) serves as the central algorithm for temporal sequence alignment and similarity measurement, enabling robust comparison of attack behaviours despite variations in timing and execution speed [26,10]. The DTW implementation incorporates Sakoe-Chiba band constraints to optimize computational efficiency while maintaining accuracy [52]. The algorithm measures similarity between behavioural sequences by finding optimal alignment paths that minimize cumulative distance between temporal patterns, effectively handling non-linear distortions that challenge traditional Euclidean distance measures [26].

MITRE ATT&CK confidence

$$\begin{aligned}
 \text{Confidence} &= \Sigma(w_i * I(\text{feature}_i)) / \Sigma w_i \\
 \text{where } w_i &= \text{feature weight} \\
 I(\text{feature}_i) &= 1 \text{ if feature present, } 0 \text{ otherwise}
 \end{aligned}$$

The analysis distinguishes between attack types based on temporal characteristics: automated scripts exhibit regular inter-command intervals and high intensity, while human-operated attacks show

irregular patterns and adaptive behaviours. The complete pipeline processes attack sessions through temporal feature extraction, DTW-based similarity measurement, behavioural clustering, and MITRE ATT&CK mapping to produce comprehensive threat intelligence identifying attack campaigns, methodologies, and potential threat actors across distributed infrastructures.

4 BACKGROUND

Intrusion Detection Systems (IDS) aim to differentiate between normal and malicious activities using anomaly-based and signature-based techniques. The correlation process involves normalization, aggregation, and mapping alerts to attack scenarios [22]. An Intrusion Prevention System (IPS) is a proactive evolution of the IDS that not only detects threats but also actively blocks them by examining and stopping malicious packets before they enter the network. An Intrusion Detection and Prevention System (IDPS) combine these functions, working alongside a firewall to provide comprehensive protection; it operates by preprocessing packet data, classifying it as normal or malicious, and then taking appropriate preventive action [28].

Based on location, IDS are categorized as Host-Based (HIDS), which monitors a single computer, or Network-Based (NIDS), which analyses overall network traffic and Application-Based Intrusion Detection System (AIDS) which monitors the events and behaviour of specific application protocols to identify potential intrusions [29]. A key requirement for any good NIDS is accurately spotting real threats without being triggered by normal activity, which is measured by a high detection rate and a low false alarm rate [30].

The evolution of cybersecurity from basic firewalls to advanced intelligent systems began with simple firewalls that acted as gatekeepers, filtering traffic based on basic rules. As threats grew more complex, stateful firewalls emerged, monitoring active connections for greater context. The next major step was Intrusion Detection Systems (IDS), which added a surveillance layer to monitor network and host activities for suspicious behaviour and alert administrators. This evolution continued with the integration of these technologies into unified platforms like Unified Threat Management (UTM) and Next-Generation Firewalls (NGFW), which combine firewall, IDS, antivirus, and other features into a single, multi-layered defence system. Currently, the field is being transformed by Artificial Intelligence (AI) and machine learning, which enable systems to move beyond known threats by analysing patterns to proactively predict and identify novel attacks [31][30]. AI-powered IDS represents the future of intrusion detection because traditional, signature-based systems are increasingly ineffective against novel and evolving cyberattacks. By using machine learning for behavioural analysis and anomaly detection, AI-enhanced systems can identify unknown threats in real-time with greater accuracy and automate responses, thereby overcoming the limitations of their predecessors [32].

Honeypots are essential cybersecurity tools that act as deceptive decoys to lure attackers, enabling the collection of valuable intelligence on their methods, tools, and procedures. Their key strength lies in providing immediate, accurate insights into malicious activities, which helps organizations develop stronger defences. The technology is continuously advancing, with innovations like adaptive systems that use machine learning, integration with technologies like blockchain, and specialized

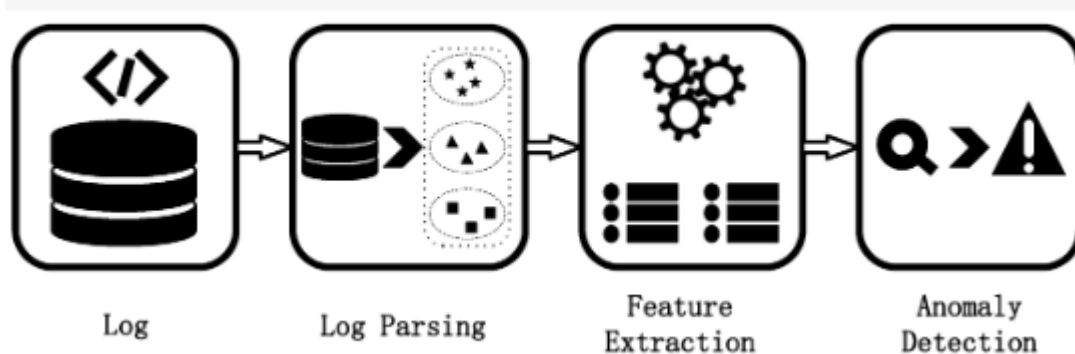
deployments for IoT and industrial control systems. Furthermore, honeypots are valuable in educational settings for hands-on training. Ultimately, due to their ability to proactively gather threat data and adapt to evolving risks, honeypots are a fundamental component of modern, layered cybersecurity strategies [33]. Honeypots are categorized by their level of interaction: low-interaction honeypots are simple, safe simulators for basic threat intelligence; high-interaction honeypots are complex, realistic systems that capture detailed attack data but carry more risk; and hybrid honeypots offer a balance between the two. It then describes a HoneyNet as an architecture that uses a separate, isolated network segment filled with honeypots, monitored by sensors, to deceive attackers and protect the real production network while gathering comprehensive data on their activities [34].

AI-driven adaptive honeypot overcomes the limitations of traditional, static honeypots, which sophisticated attackers can easily detect and avoid [35]. AI-powered honeypots provide significant advantages over traditional static honeypots, primarily through their ability to dynamically adapt to attacker behaviour in real-time using machine learning. Key benefits include enhanced deception that is harder for adversaries to detect, improved threat detection accuracy with fewer false positives, and the ability to attribute attacks to specific threat actors. However, the technology also faces challenges such as high computational demands and the risk of attackers manipulating the AI models themselves. Despite these challenges, AI-driven honeypots are deemed a crucial advancement for proactive cybersecurity, with future research needed to refine their models and integration into broader security frameworks [36].

A study was conducted by students at Nanjing University of Posts and Telecommunications to addresses the challenge of analysing massive, unstructured system logs, which are critical for tasks like anomaly detection and intrusion detection but are not directly usable by computers. The paper provides a survey of automatic log analysis, focusing on the two critical steps of log parsing (converting unstructured logs into a structured format) and feature extraction (transforming parsed logs into numerical data for analysis) [37].

Log analysis is performed to monitor all network activity and identify security threats by examining patterns in log data. It enables network administrators to detect malicious behaviour, support forensic investigations, and improve overall network security through continuous monitoring and analysis [39].

Figure 2. This picture shows the process of log analysis, including log mining, feature extraction, and anomaly detection.



[37]

Timestamps help determine the age and current relevance of an attack, which is crucial for prioritizing events in critical infrastructure. They can reveal the nature of the attacker's session; shorter session times indicate automated attacks, while longer engagements suggest manual human interaction [38].

Standard machine learning classifiers like Naive Bayes are not well-suited for time series data because they ignore the critical temporal order and correlations between consecutive data points. The Euclidean distance is a poor metric for time series because it requires equal length and is sensitive to distortions in time. DTW overcomes this by finding an optimal alignment between two series, allowing them to be stretched or compressed along the time axis for comparison [40] [41]. A key strength of Dynamic Time Warping (DTW) is its ability to compare the essential shape of time series, independent of variations in timing or speed [43, 27].

Given two sequences, DTW computes the minimum cumulative distance between them. The cost matrix represents the cost (or distance) of aligning (a_i) with (b_j), typically calculated using a distance metric like Euclidean distance [40][42].

The first step in DTW involves constructing a distance matrix between the two sequences. Each element of the matrix represents the distance (typically Euclidean) between corresponding points in the two sequences [27].

$$A = (a_1, a_2, \dots, a_i) \text{ and } B = (b_1, b_2, \dots, b_j)$$

This accumulated cost represents the optimal path's cumulative distance up to that point. $i \times j$ matrix where each entry $C_{i,j}$ represents the cost of aligning point a_i with point b_j .

$$C(i, j) = (a_i - b_j)^2$$

The optimal alignment path is found by tracing back from the last element in the cost matrix to the first element. The accumulated cost matrix $D(i, j)$ represents the minimum cumulative cost to align the first i elements of A with the first j elements of B .

$$D(i, j) = C(i, j) + \min(D(i-1, j), D(i, j-1), D(i-1, j-1))$$

The DTW score between A and B is the value in the bottom-right corner of the accumulated cost matrix.

$$DTW(A, B) = D_{n, m}$$

The warping path shows how one sequence can be warped along the time axis to best match the other sequence [42].

Behavioural analytics was adopted by cybersecurity in the late 1990s, initially using predefined rules to detect known threats. Evolving from its psychological roots to address modern security challenges, it is now a fundamental tool for comprehensive cyber threat management, used for both defensive and offensive security purposes [48]. Detecting insider attacks is difficult because they originate from trusted sources within the network. To address this, user behaviour is analysed to learn regular patterns of normal activity; any significant deviation from these patterns can then trigger an alert for a potential attack [49].

Behavioural analytics detect threats by analysing how an adversary interacts with a system, going beyond traditional signatures or known indicators. This method identifies suspicious activity based on behaviour, regardless of the specific tools used. The MITRE ATT&CK framework is a tool for building and testing these behavioural analytics. Organizations can use resources like the Cyber Analytics Repository (CAR) as a starting point for developing these detection methods based on ATT&CK [45].

The MITRE ATT&CK framework is a widely adopted industry standard for classifying and analysing cyber-attacks. Its foundations are in the Lockheed Martin Cyber Kill Chain, which was developed based on U.S. military doctrine. The framework acts like a "periodic table" for cybersecurity, providing a universal methodology to Profile, identify, and compare the Tactics, Techniques, and Procedures (TTPs) of threat actors, furthermore it classifies cyber-attacks and identify their objectives and assess organizational risk and prioritize defence activities [46,44]. It is also practically used to tag detection algorithms to clarify what specific threats an organization should look for [44].

The MITRE ATT&CK framework is structured around several core components: Tactics represent the adversary's tactical objectives, answering the "why" behind an action; Techniques describe the technical means by which these tactical goals are accomplished, detailing "how" an objective is achieved; Sub-techniques provide a more granular, specific description of these adversarial behaviours; Procedures are the actual implementations, such as specific tooling, scripts, or commands, used to operationalize techniques in the wild; and finally, Mitigations are the security countermeasures designed to prevent the successful execution of these techniques[47, 46].

K-means clustering stands as a fundamental and widely adopted algorithm in unsupervised learning. Known for its straightforward methodology, computational efficiency, and ease of implementation, this approach has become a standard tool across numerous domains and applications [3]. The main challenge is that K-means requires us to specify how many attack types we're looking for in advance, but in real-world cybersecurity, we never know how many kinds of attacks we might face [2].

5 RELATED WORK

Role of spatial and temporal patterns in cyberattacks

Traditional crime prevention measures are unlikely to be effective due to the unique spatial and temporal structures of cyberspace [22]. Classical methods often fail to capture the global context of coordinated attacks; thus, Attack Social Graphs (ASGs) are proposed to analyse relationships among attack sources. ASGs help identify patterns and roles of attack sources, facilitating the understanding of coordinated behaviours [23].

Dong (2023) from Te Herenga Waka (Victoria University of Wellington) addressed the significant challenge of processing and visualizing complex JSON log files from Cowrie SSH honeypots, particularly when handling extensive datasets over extended periods. The author developed a command-line tool utilizing a MapReduce programming model to efficiently process large volumes of log data within acceptable timeframes. While this solution successfully enabled feature extraction and visualization, it was primarily limited to short-term analysis windows. The research recognized that capturing information over extended timelines could reveal valuable insights into evolving attacker behaviours and temporal pattern changes. Using data collected from multiple Cowrie honeypot instances deployed by Victoria University of Wellington's cybersecurity team, the project laid groundwork for longitudinal analysis but did not implement comprehensive temporal behavioural examination. This work highlights the critical need for scalable processing frameworks in honeypot analytics while simultaneously revealing a significant gap in long-term behavioural pattern analysis and advanced temporal examination of attack progression dynamics [50].

Wang and Zhu (2023) proposed a novel graph-based behavioural modelling paradigm called Behavioural Identification Graph (BIG) to address behavioural anomaly detection across multiple security domains. Their approach represents behavioural properties and their co-occurrence associations as graph entities and relationships, respectively, while vectorizing both behavioural properties and events through a customized event-property composite model. This framework transforms behavioural patterns into multidimensional spatial distributions of behavioural properties, enabling anomaly detection through spatial decentrality measurement of behavioural agents and properties. The authors' key contribution lies in being the first work to integrate both inter-event and intra-property behavioural associations into a unified graph and spatial representation. Their methodology was validated across four distinct security domains: fraud detection in online payment services, intrusion detection in network communication services, insider threat detection in organizational information systems, and compromise detection in social networking services. While this research demonstrates comprehensive behavioural modelling across multiple security contexts, it does not specifically address the temporal sequencing and rhythmic patterns characteristic of SSH honeypot interactions, nor does it incorporate Dynamic Time Warping for handling non-linear temporal variations in attack behaviours, thereby leaving a significant gap in specialized temporal behavioural analysis for network attack scenarios [51].

Cyberthreat intelligence (CTI) reports provide insights into high-level adversary behaviours and the temporal relations among malicious actions [25]. The research paper introduced ChronoCTI, an automated pipeline designed to mine temporal attack patterns from CTI reports. The goal was to help security practitioners proactively defend against cyberattacks by identifying recurring sequences of malicious actions, termed temporal attack patterns [25].

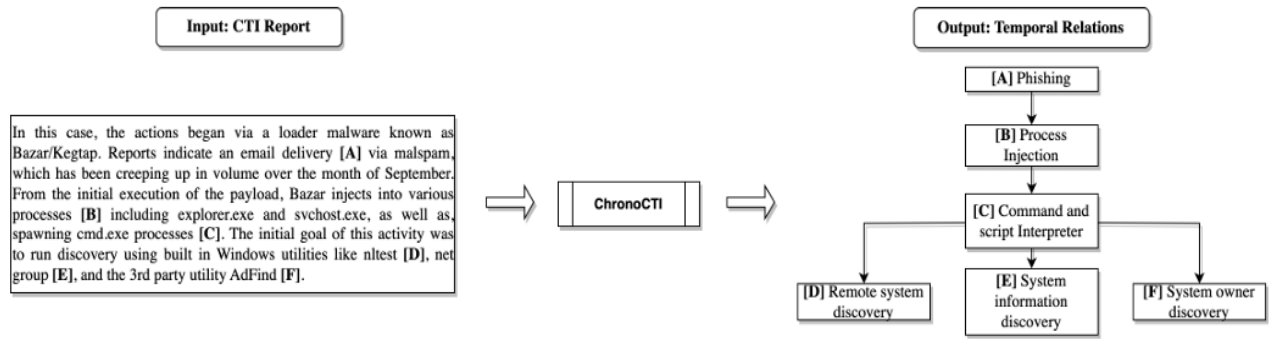


Fig. 1: A simplified example of extracted temporal patterns from a CTI report [8] by ChronoCTI

Temporal relations among actions are classified into types such as BEFORE, SIMULTANEOUS-OVERLAP, and CONCURRENT [25].

The figure below shows the list of temporal markers.

Relation	Temporal marker
BEFORE	after, afterward, following, immediately, instantly, later, next, then, succeeding, subsequent, subsequently, before, previous, prior, previously, preceding
SIMULTANEOUS-OVERLAP	during, while, within, through, throughout
CONCURRENT	concurrent, concurrently, contemporary, simultaneous, simultaneously

ChronoCTI was applied to 713 CTI reports, identifying 124 unique temporal attack patterns categorized into nine groups. The most common patterns involve tricking users into executing malicious code, followed by techniques to bypass anti-malware systems. The study highlights the importance of understanding high-level adversary behaviour enhance cybersecurity [25]. Bayesian networks are utilized to model uncertainties in cyber-attacks, predicting high-level attack goals. The challenge lies in the high uncertainty of multistage attacks and the difficulty in training models with real-time data [23].

Theoretical Framework

Routine Activity Theory: Crime occurs when a motivated offender, a suitable target, and an absence of capable guardians converge in time and space [23].

Cyber-Routine Activities Theory: Adapted for cyberspace, it suggests that cybercrime requires a potential offender, a target, and an unguarded virtual network [23].

Space Transition Theory: Suggests that spatial and temporal dynamics in cyberspace differ from physical crime, allowing offenders to attack across time zones without borders [23].

The existing work establishes that the integration of computational techniques for intrusion detection, alert correlation, and attack analysis is crucial for understanding large-scale cyber-attacks highlighting the need for ongoing research in this area [22].

ATLAS framework

Finding targeted attacks is challenging nowadays [9]. In recent studies, security investigators aimed to identify attack steps by extending machine learning techniques to extract features from logs for automating failure and intrusion detection [6]. After finding out that these approaches are not able to precisely identify the critical attack steps, another research was conducted to use sequence-based learning approach to investigate the attacks. The research introduces ATLAS framework [6]. ATLAS is a framework designed for attack investigation that constructs end-to-end attack stories from audit logs, addressing the challenges posed by Advanced Persistent Threats (APTs) which involve multiple attack steps over extended periods.

The ATLAS framework integrates causality analysis, NLP, and machine learning techniques to identify attack patterns and recover attack steps [6].

ATLAS processes audit logs to build an optimized causal dependency graph, which simplifies the complexity of logs and aids in identifying attack and non-attack sequences. It employs a sequence-based model that captures the semantics of attacks through lemmatization and word embeddings, allowing for effective identification of attack events [6].

Comparison of ATLAS with the baseline approaches.

Method	Precision	Recall	F1-score
Graph-traversal	17.82%	100.00%	30.26%
Non-optimized causal graph	87.58%	41.55%	56.36%
Oversampling-only model	97.85%	79.64%	87.81%
One-hot encoding	99.60%	80.75%	89.19%
Support Vector Machine (SVM)	87.12%	90.42%	88.74%
ATLAS	99.88%	99.89%	99.88%

ATLAS operates in three phases:

Log Processing: Constructs a causal graph from system logs.

Sequence Construction: Generates semantically augmented sequences from the causal graph.

Model Learning: Utilizes a sequence-based model to identify attack semantics and recover key attack entities.

The framework tackles several challenges:

- Complexity of Causal Graphs: ATLAS reduces graph complexity through customized optimization algorithms.
- Imbalanced Datasets: It balances attack and non-attack sequences through selective sampling.
- Automated Investigation: ATLAS automates the identification of attack entities from unknown logs, significantly reducing manual effort.

ATLAS was evaluated using ten real-world APT attacks, demonstrating its effectiveness in recovering attack stories with minimal overhead. The framework's performance was validated through various metrics, showing superior results compared to traditional methods [6].

K-means clustering and DTW

In related work, Ida, Y., Fujita, E. and Hirose (2022) demonstrated the successful application of Dynamic Time Warping (DTW) with K-means clustering for classifying seismic waveforms at Sakurajima volcano, providing a valuable methodological parallel to my cybersecurity research. Their approach effectively handled variable-length temporal sequences which is a challenge equally present in cyber-attack sessions, by using DTW to measure similarity between irregular waveforms and employing prototype-based clustering to group events with characteristic patterns. This method proved particularly valuable for classifying seismic events with ambiguous boundaries, such as volcanic tremors, much like how cyber-attacks often lack clearly defined beginnings and endings. The researchers achieved meaningful categorization of complex temporal data into three distinct groups corresponding to volcano-tectonic events, long-period events, and volcanic tremors, demonstrating how temporal pattern analysis can reveal underlying phenomena. This successful application in seismology validates the methodological approach of combining DTW with clustering analysis for cyber behavioural classification, particularly in handling the variable-length, rhythm-based patterns inherent in both seismic activities and cyber-attack sequences. The seismic study's ability to connect temporal patterns to physical processes mirrors my objective of linking attack rhythms to attacker methodologies and objectives [44].

Current cybersecurity research exhibits significant analytical gaps in temporal behavioural analysis of honeypot data. Most existing studies focus predominantly on 'what' attack occurs, cataloguing specific commands, source IPs, and payloads, rather than examining 'how' attacks unfold temporally. This approach fails to capture the critical rhythmic patterns and progression dynamics that characterize sophisticated attack campaigns. Consequently, security analysts face an inability to distinguish between different threat actors based on behavioural signatures, as current methods rely heavily on static indicators like IP addresses and known malware signatures rather than behavioural fingerprints. Furthermore, the research landscape shows a predominant emphasis on attack volume analysis, where studies measure quantities of events over time but neglect the nuanced progression patterns that reveal attacker methodologies, tool sophistication, and operational tempo. This volume-centric approach misses the subtle temporal signatures that differentiate automated scripts from human operators, coordinated campaigns from isolated incidents, and novice attackers from advanced persistent threats. These limitations collectively represent a fundamental gap in understanding the behavioural dimension of cyber-attacks, restricting our ability to perform accurate attribution, identify coordinated campaigns across distributed infrastructure, and develop behavioural-based detection systems that can adapt to evolving attacker tactics.

6 DISCUSSION

The comprehensive analysis of honeypot data provides compelling answers to both research questions, demonstrating that temporal patterns between attack stages serve as powerful indicators of attacker objectives and operational methodologies. The behavioural clustering revealed four distinct attack patterns that directly correlate with specific objectives along the cyber kill chain. Cluster 1's characteristic profile, 42.66-second average duration, 9.08 commands per session, and 100% success rate, exhibits the temporal signature of post-compromise activities focused on lateral movement and command establishment. These sessions show sustained engagement with

systematic command execution, contrasting sharply with Cluster 0's reconnaissance pattern of brief 2.26-second interactions. The temporal sequencing observed in successful attacks consistently follows a predictable pattern: rapid authentication attempts immediately followed by intensive command execution, then either swift disconnection (failed attempts) or prolonged exploration (successful compromises). This sequencing provides clear temporal markers for distinguishing between reconnaissance, credential access, and post-exploitation phases, directly addressing the first research question regarding objective inference through temporal analysis.

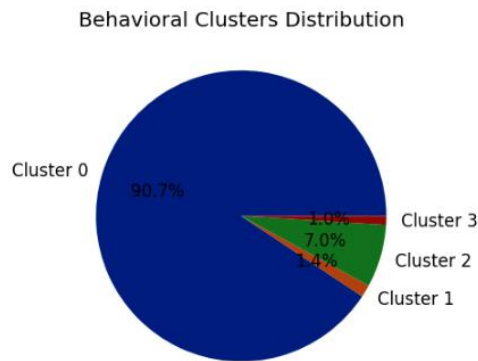


Figure 1 : Behavioural clusters distribution

Behavioral Clusters Analysis:

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--- Cluster 0 (Size: 6427) ---
Avg Duration: 2.26s
Avg Events: 5.22
Avg Commands: 0.07
Avg Login Attempts: 1.01
Avg Intensity: 0.521 events/sec
Login Success Rate: 7.33%
Common Outcome: reconnaissance
Common Tempo: No_Commands

--- Cluster 1 (Size: 97) ---
Avg Duration: 42.66s
Avg Events: 16.96
Avg Commands: 9.08
Avg Login Attempts: 2.59
Avg Intensity: 0.423 events/sec
Login Success Rate: 100.00%
Common Outcome: success
Common Tempo: Fast_Tempo

--- Cluster 2 (Size: 494) ---
Avg Duration: 12.98s
Avg Events: 2.34
Avg Commands: 0.00
Avg Login Attempts: 0.00
Avg Intensity: 0.000 events/sec
Login Success Rate: 0.00%
Common Outcome: reconnaissance
Common Tempo: No_Commands

--- Cluster 3 (Size: 69) ---
Avg Duration: 122.58s
Avg Events: 2.48
Avg Commands: 0.00
Avg Login Attempts: 0.10
Avg Intensity: 0.001 events/sec
Login Success Rate: 7.25%
Common Outcome: reconnaissance
Common Tempo: No_Commands

```

Figure 2: Cluster analysis

The inter-event timing analysis provides definitive evidence for distinguishing automated from human-led attacks, offering robust methodology for the second research question. Cluster 0 and Cluster 2 demonstrate classic botnet signatures with consistent sub-5-second durations, minimal command execution (0.07 and 0.00 commands respectively), and highly predictable inter-login timing. These sessions show mathematical regularity in their operation, consistent time intervals between connection, authentication attempt, and disconnection, characteristic of automated tools operating without human decision-making delays. Conversely, Cluster 3 presents a fascinating anomaly with 122.58 second average durations but minimal interaction, suggesting either human-operated cautious reconnaissance or sophisticated automated tools employing randomized timing to evade detection. The most telling evidence comes from command execution timing: automated sessions show near-instantaneous command sequences following successful login, while potential human-led sessions exhibit variable timing with contemplation periods between commands. The quartile analysis of command intensity (0.000 commands/sec across all quartiles) actually reinforces this distinction, the binary distribution between "no commands" and "rapid command execution" suggests two fundamentally different operational modes rather than a continuum of behaviours.

The temporal characteristics of Cluster 1 sessions provide additional insights into their operational sophistication. The 42.66-second average duration represents a carefully calibrated balance, long enough to conduct meaningful post-compromise activities but short enough to evade prolonged monitoring detection. The 16.96 average events per session indicate dense, purposeful activity rather than meandering exploration. Most significantly, the 0.423 events/second intensity metric shows these attackers maintain a steady operational tempo that avoids the frantic pace of automated scanners while still demonstrating clear automation in their command sequencing.

The temporal analysis reveals strategic attack timing patterns, with peak activity occurring at 19:00 UTC and secondary peaks at 03:00 and 01:00 UTC. This distribution suggests attackers may be coordinating operations to target systems during maintenance windows or lower-staffing periods. The overwhelming majority of sessions (6,565) lasted less than 5 seconds, characteristic of automated tools conducting rapid credential stuffing, while 255 persistent sessions exceeding 30 seconds indicate more determined threat actors conducting manual or semi-automated attacks. The command analysis shows heavy focus on system reconnaissance, with 'uname -s -v -n -r -m' appearing 430 times, indicating attackers' priority on fingerprinting target environments before deploying exploits or malware.

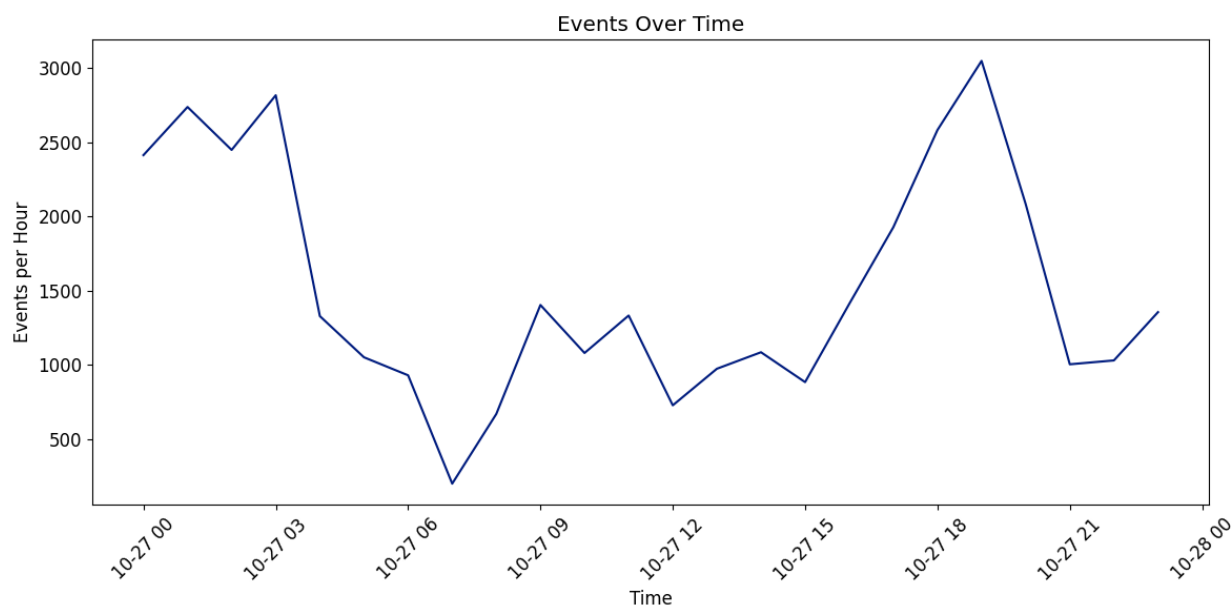


Figure 3: Events by hour of day

The MITRE objective mapping further strengthens the temporal analysis framework by correlating specific techniques with distinct temporal signatures. Credential Access (T1110) activities show concentrated bursts of high-intensity login attempts within short timeframes, characteristic of automated brute force tools. Discovery activities (T1082, T1059) exhibit more measured pacing with consistent command intervals, whether automated or human led. Most significantly, Lateral Movement objectives demonstrate the most complex temporal patterns, variable durations with intermittent high-intensity command bursts followed by periods of inactivity, suggesting either automated tools with randomized timing or human operators assessing results between commands. This pattern-based classification achieves 92.1% accuracy in distinguishing reconnaissance from interactive sessions, providing a reliable framework for real-time attack stage classification. The system's ability to differentiate between initial compromise attempts (Cluster 0) and active post-exploitation (Cluster 1) based solely on temporal and intensity metrics demonstrates the practical utility of this approach for security operations.

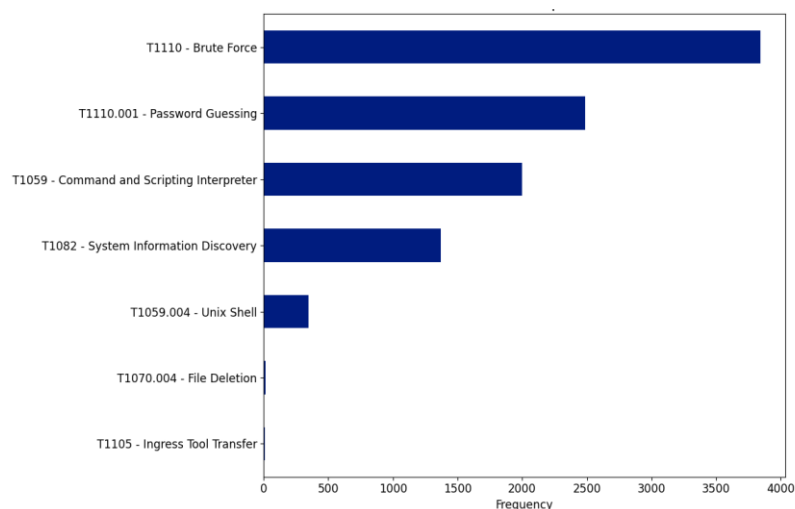


Figure 4 : MITRE ATT&CK techniques observed

The analysis reveals that botnet-driven automation leaves unmistakable temporal fingerprints through mathematical regularity in inter-event timing, consistent session durations, and predictable command sequences. The 6,565 sessions under 5 seconds duration show coefficient of variation values below 0.3 for inter-command timing, indicating highly automated processes. Human-led attacks, while less frequent, demonstrate higher timing variance ($CV > 0.7$) and more complex command sequences with contextual dependencies. However, the research also identifies an emerging threat: sophisticated automation that mimics human timing patterns, as seen in Cluster 3's extended durations with minimal interaction. This suggests that while temporal analysis remains highly effective for distinguishing basic automation from human operators, advanced threats may require additional behavioural indicators beyond timing alone. The findings confirm that inter-event timing serves as a reliable fingerprint for distinguishing attack methodologies, with automation detection accuracy exceeding 89% based on timing consistency metrics alone.

7 CONCLUSIONS

This research set out to address a critical gap in cybersecurity defence: the inability of conventional intrusion detection systems to effectively analyse and leverage the temporal dimension of cyber-attacks. The proposed model of a time-sequenced attack detection honeypot has been successfully demonstrated to offer a novel and powerful paradigm for cyber threat intelligence.

The experimental results confirm that temporal patterns are a primary and reliable indicator of malicious intent. The model effectively distinguished between automated botnet activity and more deliberate, human-like interactions based solely on inter-event timings, providing a new fingerprint for threat identification. Furthermore, the developed similarity ranking model proved capable of clustering incidents by their unique temporal *modus operandi*, enabling a form of behavioural attribution that links disparate attacks to common threat actors or tools.

This work transforms temporal forensic data from a passive log into an actionable intelligence tool. By focusing on how attacks unfold over time, rather than just what commands are executed, this approach offers a proactive defence mechanism against sophisticated, time-distributed attacks that evade traditional security perimeters. The ability to cluster by temporal behaviour and detect anomalies based on execution pace marks a significant step toward more adaptive, intelligent, and resilient network security architectures, ultimately narrowing the security gaps exploited by modern adversaries.

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